

Study design and analysis pipeline

A Dataset & cohorts

spine-generic-multi-subject · open BIDS

267 subjects

42 sites · 13 scanner models

HC cohort
 $N = 188$

ALL cohort
 $N = 246$
(HC + MildComp + DCM)

Vendor composition (n)



B Biomarker extraction at C2-C3

Spinal Cord Toolbox v6.4 · 8 per-subject scalars

T2w	T2*	MT	DTI
		MTR	FA
Cord CSA	GM CSA	MTsat	MD
			AD
			RD

C Harmonization methods

ordered by modeled vendor components

LME	ComBat	ComBat-joint	CovBat	RELIEF
mean only (random intercept)	+ scale (per-metric, EB)	+ joint EB (across metrics)	+ covariance (PCA + ComBat)	+ latent factors (ICA + BH-FDR)

Increasing model complexity

D Evaluation framework

three orthogonal endpoints

Vendor removal	η^2 · ICC · PERMANOVA R^2
Biology recovery	age R^2 · sex R^2
Subject preservation	within-vendor $r_{\text{pre, post}}$